DIMENSIONS OF LOCAL NOETHERIAN SPACES ADD ALONG A FLAT MORPHISM OF RELATIVE DIMENSION D

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries the hypotheses flat, locally of finite type and relative dimension d; all of them are inside the counted statement span.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Dimension of an algebraic space at a point. Properties of Spaces, Definition spaces-properties-definition-dimension-at-point, used at proof lines 6843-6845. Gives the meaning of the two dimensions in the formula.
- The definition of relative dimension d. used at proof lines 6845-6846. Relates the fibre dimension to d.
- The scheme case for the dimension formula. Morphisms, Lemma morphisms-lemma-rel-dimension-dimension, used at proof lines 6847-6848. Is the input that closes the argument.

Source reference. Original lemma 0AFE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13157–13157. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 04N5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6066–6066. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally Noetherian algebraic spaces over S which is flat, locally of finite type and of relative dimension d. For every point x in $|X|$ with image y in $|Y|$ we have* $\dim_x(X) = \dim_y(Y) + d$.

Proof. By definition of the dimension of an algebraic space at a point (Properties of Spaces, Definition 04N5) and by definition of having relative dimension d, this reduces to the corresponding statement for schemes (Morphisms, Lemma 0AFE). □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.